



Computational Analysis of Non-coding DNA Regions: Understanding the Role of Encode Candidate *cis*- Regulatory Elements in Cancer and Gene Regulation

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Abstract

This study investigates the role of candidate cis-regulatory elements (cCREs) in cancer by integrating ATAC-seq data, ENCODE cCRE annotations, and TCGA copy number variation (CNV) datasets. Reproducing previous findings, we confirmed that most cancer-specific ATAC-seq peaks overlap with ENCODE cCREs, particularly enhancer and promoter elements. The study developed a cancer cCRE Atlas, revealed cross-cancer regulatory similarities and differences, and discovered a correlation between CNVs and chromatin accessibility. Tumor-specific cCREs were linked to oncogenic genes and pathways. These insights advance understanding of the epigenetic regulation of cancer and highlight novel therapeutic targets.

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1. Introduction

Cancer progression is fundamentally linked to disruptions in gene regulation, a process that determines when, where, and how genes are expressed. Gene regulation is tightly controlled in normal cells to ensure proper cellular function. However, in cancer cells, these regulatory mechanisms are altered, leading to the activation of oncogenes that drive cell growth and survival or the inactivation of tumor suppressor genes, resulting in uncontrolled proliferation. These changes are often mediated through modifications in candidate cis-regulatory elements (cCREs), including promoters, enhancers, and silencers, which serve as critical modulators of gene expression.

Chromatin accessibility plays a central role in this process by controlling which parts of the genome are available for transcription. Accessible regions of chromatin, marked by specific histone modifications and chromatin remodeling, allow transcription factors and other regulatory proteins to bind DNA and initiate gene expression. Conversely, inaccessible chromatin restricts this activity. Dysregulation of chromatin accessibility indicates cancer, leading to abnormal gene expression patterns that promote tumorigenesis. Advanced sequencing technologies, such as ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing), have provided powerful tools to explore these changes, enabling the identification of open chromatin regions that may serve as active regulatory elements.

The ENCODE (Encyclopedia of DNA Elements) project has significantly advanced our understanding of gene regulation by cataloging many cCREs across the human genome. By integrating ATAC-seq data from different types of cancers with the Registry of cCREs developed by ENCODE, researchers can identify how cancer cells exploit these regulatory elements to drive oncogenesis. The Moore Lab at UMass Chan Medical School has applied this approach to investigate chromatin accessibility across multiple cancer types, offering insights into the epigenetic mechanisms underlying cancer progression.

This study seeks to understand the role of cCREs in cancer development, including their function, distribution, overlap between 23 cancer types, and comparison to healthy tissues. This research seeks to uncover the role of epigenetic changes in cancer and explore potential

therapeutic targets within dysregulated regulatory elements. These findings will contribute to a growing body of knowledge that links chromatin dynamics and cCREs to cancer biology, highlighting opportunities for innovative treatments that target the epigenome.

1.1 Gene Regulation and Its Role in Cancer Progression

Gene regulation is a fundamental process that controls gene expression, controlling when and where specific genes are activated or silenced. This regulation is precise in normal cells, ensuring that genes are expressed correctly to maintain cellular function. In cancer, this regulation is disrupted, leading to the activation of oncogenes. Activated oncogenes promote cell growth and survival or the inactivation of tumor suppressor genes, leading to uncontrolled cellular proliferation.

cCREs such as promoters, enhancers, and silencers are critical to gene regulation. These regulatory elements interact with transcription factors, which are proteins that bind to specific DNA sequences to either promote or inhibit gene expression. In cancer, mutations or epigenetic alterations in these regulatory elements can lead to abnormal gene expression. (Grandi, Modi et al. 2022, Zhang, Wang et al. 2024). For instance, in T-cell acute lymphoblastic leukemia (T-ALL), a somatic mutation creates a novel binding site for the MYB transcription factor in an enhancer upstream of the TAL1 oncogene. This aberrant enhancer activation leads to overexpression of TAL1, promoting leukemogenesis (Mansour, Abraham et al. 2014). Similarly, tumor suppressor genes can be silenced due to changes in chromatin accessibility or DNA methylation at regulatory sites, particularly within CpG islands. CpG islands are regions of the genome with a high frequency of cytosine-phosphate-guanine (CpG) dinucleotides, commonly located at or near gene promoters. In normal cells, these CpG islands are typically unmethylated, which allows for active transcription of the associated gene. However, in cancer, aberrant hypermethylation of CpG islands within promoter regions can repress gene expression by preventing transcription factor binding and recruiting proteins that condense chromatin structure. For example, hypermethylation of the CpG island at the CDKN2A promoter region is frequently observed in various cancers and leads to silencing of this critical tumor suppressor gene (Fig. 1) (Esteller 2002). This epigenetic silencing contributes to tumorigenesis by disabling key pathways that control cell proliferation and apoptosis.

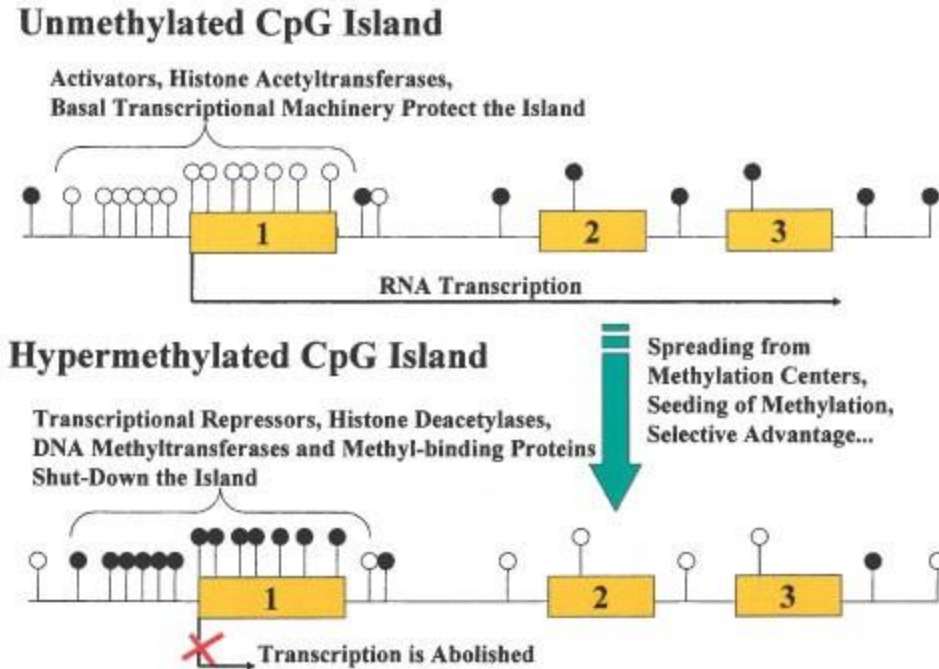


Figure 1: CpG island of a tumor suppressor gene is represented in a normal and a tumor cell (Esteller 2002)

The presence of a dense hypermethylation changes completely its molecular environment. White dots, unmethylated CpGs; Black dots, methylated CpGs (Esteller 2002).

Chromatin accessibility is critical in determining which genes are actively expressed in a cell. In regions where chromatin is “open,” transcription factors and other regulatory elements can easily access the DNA, allowing transcription to occur. In contrast, “closed” chromatin is tightly packed, preventing access to the enclosed DNA. Various chromatin remodelers and histone modifications regulate the accessibility of chromatin.

Chromatin accessibility is frequently altered in cancer. Using techniques such as ATAC-seq, researchers can determine the regions of a genome that remain accessible or become newly accessible in cancer cells. These accessible regions often correspond to enhancers or promoters of genes critical for cancer progression, such as those involved in cell proliferation, survival, or metastasis. By understanding this network of chromatin regions, researchers can identify regulatory elements hijacked by cancer cells to promote tumor growth. (Hartl, Zhuang et al. 2024, Zhang, Wang et al. 2024).

1.2 ATAC-seq Technology and Its Role in Cancer Research

ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing) profiles chromatin accessibility across the genome. Introduced in 2013, it has significantly advanced our understanding of gene regulation by enabling the identification of open chromatin regions, which are indicative of active regulatory elements such as promoters, enhancers, insulators, and other cis-regulatory elements. Open chromatin regions are accessible to transcription factors and other regulatory proteins, making them critical to the control of gene expression.

The method uses a hyperactive Tn5 transposase enzyme, which cleaves accessible chromatin and inserts sequencing adapters into these regions. This targeted action ensures that only the regions of DNA exposed and accessible in the chromatin landscape are tagged and amplified for sequencing. The resulting sequencing data highlights accessible regions in chromatin. (Grandi, Modi et al. 2022)

A key feature of ATAC-seq is its scoring system, which quantifies chromatin accessibility. Peaks in the ATAC-seq data represent regions of open chromatin, and their intensity reflects the level of accessibility. These scores are calculated based on the density of sequencing reads mapped to a specific region. High-intensity peaks correspond to highly accessible chromatin regions, often associated with active regulatory elements. By comparing these peaks across different conditions or cell types, researchers can infer changes in chromatin accessibility, which correlate with alterations in gene regulation.

ATAC-seq has become an indispensable tool for understanding the epigenetic changes that drive tumorigenesis in cancer research. Chromatin accessibility profiles generated by ATAC-seq reveal critical regulatory elements differentially active in cancer cells. For instance, the technique has been used to identify oncogene enhancers that become hyperactive and tumor suppressor regions that lose accessibility. These changes frequently arise from epigenetic reprogramming or chromatin remodeling, making ATAC-seq invaluable for studying the molecular mechanisms of cancer. (Corces, Granja et al. 2018, Hartl, Zhuang et al. 2024). Below is the schematic of a typical ATAC-seq peak calling analysis (Fig. 2).

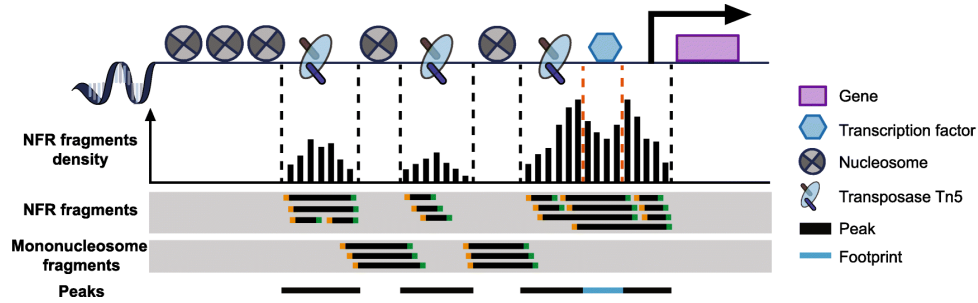


Figure 2: ATAC-seq peak calling analysis (Yan, Powell et al. 2020)

Schematic of ATAC-seq data from core analyses. In an ATAC-seq experiment, Tn5 binds and cuts open chromatin and simultaneously ligates adapters. The fragments are sequenced to identify open chromatin regions (black) and footprints (blue). NFR fragments represent the open chromatin, while nucleosome-bound fragments reflect nucleosome positions (gray shaded tracks) (Yan, Powell et al. 2020)

1.3 The ENCODE Project and the Registry of cCREs

The ENCODE (Encyclopedia of DNA Elements) project, initiated in 2003, is a significant resource for cancer research. It aims to identify and catalog all functional elements in the human genome, including protein-coding regions, non-coding RNAs, and regulatory elements. One of its key contributions is the identification of cCREs, non-coding DNA regions that control the transcription of nearby genes by serving as binding sites for transcription factors and other regulatory proteins. The comprehensive data provided by the ENCODE project has been instrumental in understanding the role of cCREs in cancer progression.

The Registry of cCREs, developed in phase three of the ENCODE project, is a centralized database of all identified cCREs, serving as a resource for researchers studying gene regulation. The registry, which includes 926,535 human cCREs and 339,815 mouse cCREs, was developed using data from ATAC-seq, histone profiling, and transcription factor binding. cCREs are designated by their regulatory function, such as promoter or enhancer functionality. (Abascal, Acosta et al. 2020)

In cancer research, the Registry of cCREs helps identify regulatory elements that may be repurposed by cancer cells. Many regulatory elements active in normal cells are used by cancer cells to promote the expression of oncogenes or silence tumor suppressors. By comparing the chromatin accessibility profiles of cancer cells found using ATAC-seq with cCREs cataloged

by ENCODE, researchers can identify which regulatory elements are active in cancer and potentially drive tumorigenesis. (Hartl, Zhuang et al. 2024, Zhang, Wang et al. 2024).

1.4 cCRE Classes

In the ENCODE project, cCREs are classified into several categories based on their biochemical signatures and roles in gene expression (Fig. 3). These categories help researchers understand how genome regions control gene expression across various cell types.

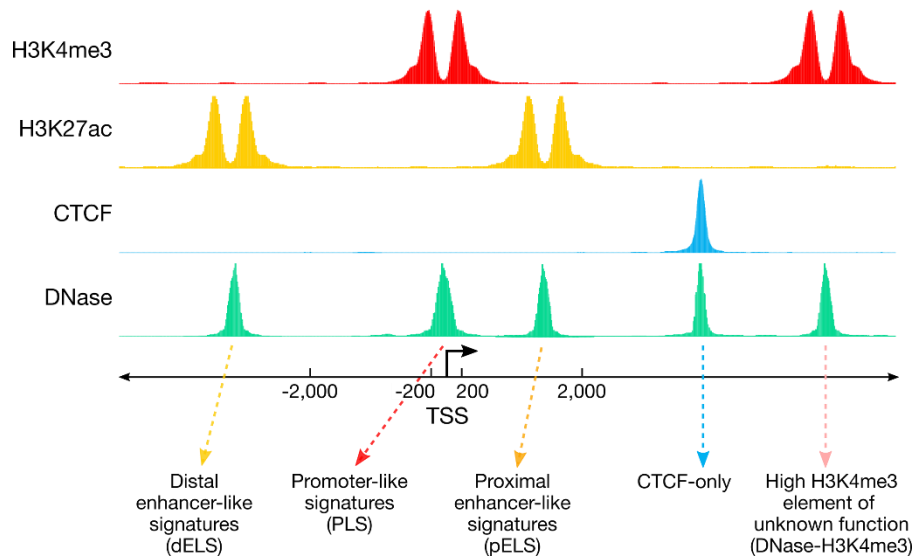


Figure 3: Chromatin signatures and classification of cCREs

Chromatin profiles for H3K4me3, H3K27ac, CTCF, and DNase are displayed relative to the transcription start site (TSS), illustrating distinct cis-regulatory element (cCRE) types. DNase peaks mark distal enhancer-like signatures (dELS) without high H3K4me3 levels. Promoter-like signatures (PLS) feature strong H3K4me3 and H3K27ac signals near the TSS. Proximal enhancer-like signatures (pELS) exhibit DNase and H3K27ac peaks closer to the TSS. CTCF-only elements show distinct CTCF and DNase signals. High H3K4me3 elements of unknown function are defined by DNase and H3K4me3 co-localization. (Consortium 2024)

1.4.1 Promoter-like cCREs (PLS)

These elements are marked by H3K4me3, a histone modification associated with transcription start sites and accessible chromatin. As shown in Figure 3, the peak for H3K4me3 is centered near the transcription start site (TSS), indicating active promoter regions that initiate gene transcription. Promoters are vital for initiating gene transcription, and the cCREs in this class often include elements that regulate the expression of nearby genes. (Abascal, Acosta et al. 2020)

1.4.2 Enhancer-like cCREs

Proximal Enhancer-like Signatures (pELS) are near transcription start sites (TSS). They are characterized by specific chromatin features such as DNase hypersensitivity and H3K27ac, a histone modification associated with active enhancers. DNase hypersensitivity is identified through DNase-seq, an assay that uses DNase I enzyme to cleave accessible chromatin regions, revealing open and regulatory areas. H3K27ac enrichment is assessed using chromatin immunoprecipitation followed by sequencing, ChIP-seq, which isolates DNA bound to the H3K27ac modification, indicating enhancer activity. Figure 2 illustrates these pELS near the TSS, showing significant peaks in H3K27ac and DNase hypersensitivity signals. (Abascal, Acosta et al. 2020)

Distal Enhancer-like Signatures (dELS) are elements located farther from the TSS. They are similarly marked by H3K27ac and DNase peaks, but they regulate gene expression from a distance, affecting gene transcription in different ways depending on tissue context. (Abascal, Acosta et al. 2020)

1.4.3 CTCF-only cCREs

These elements are indicated by CTCF, a highly conserved zinc finger protein that organizes chromatin structure and establishes insulation between regulatory domains. CTCF binds to specific DNA sequences to mediate chromatin looping and create boundaries that prevent improper interactions between promoters and enhancers. As shown in Figure 2, CTCF-only elements have distinct peaks for CTCF binding sites and serve this organizational function, ensuring proper gene regulation and genome stability. (Abascal, Acosta et al. 2020).

1.4.4 DNase-H3K4me3 cCREs

These regions show strong DNase hypersensitivity and high levels of H3K4me3 but have no associated H3K27ac signal, suggesting they may serve unknown regulatory roles. They are depicted as having distinct peaks for H3K4me3 and DNase without additional enhancer-related histone marks. (Abascal, Acosta et al. 2020)

1.4.5 Why is ATAC-seq Used Over Other Methods

ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing) is the optimal technique for comparing DNA accessibility in cancer with ENCODE-defined cCREs due to its

high sensitivity, resolution, and unbiased genome-wide approach. Unlike ChIP-seq, which targets specific histone modifications or transcription factors, ATAC-seq directly measures chromatin accessibility, capturing changes across promoters, enhancers, and insulator regions. Its low input requirements make it ideal for cancer research, where samples may be limited, and it efficiently aligns accessible chromatin regions with ENCODE's cCRE classifications. By mapping cancer-associated chromatin remodeling to these regulatory categories, ATAC-seq provides insights into dysregulated elements driving aberrant gene expression in cancer.

1.5 Comparison of ATAC-seq Data and cCREs by Corces et al.

Comparing ATAC-seq data with the Registry of cCREs from the ENCODE project is essential in understanding how regulatory elements are involved in cancer gene regulation. ATAC-seq identifies regions of accessible chromatin, which typically correspond to active regulatory elements. On the other hand, the cCREs cataloged by ENCODE provide a comprehensive reference of regulatory elements across a wide range of human tissues and cell types. (Pundhir 2015). By comparing the chromatin accessibility profiles generated by ATAC-seq with the cCREs, researchers can determine which known regulatory elements are utilized in cancer cells. (Corces, Granja et al. 2018, Zhang, Wang et al. 2024).

Corces et al. (2018) utilized ATAC-seq to characterize the chromatin accessibility landscapes of 410 primary tumor samples spanning 23 cancer types from The Cancer Genome Atlas (TCGA). Their study identified 562,709 accessible DNA regulatory elements, greatly expanding cancer's known catalog of cis-regulatory elements.

Through integration with TCGA multi-omic datasets, the authors linked chromatin accessibility profiles to cancer-specific regulatory networks. They identified many distal regulatory elements interacting with gene promoters, including those driving the expression of key oncogenes such as MYC, SRC, and PDL1, a target for cancer immunotherapy. These findings highlighted the critical role of transcription factors (TFs) in modulating cancer-specific gene expression and revealed noncoding somatic mutations that alter chromatin to drive tumor progression. For example, a single-nucleotide mutation upstream of the FGD4 gene created a de novo binding site for an NKX transcription factor, increasing chromatin accessibility and gene expression.

The study also demonstrated the utility of chromatin accessibility profiling in defining cancer subtypes, identifying novel prognostic markers, and uncovering regulatory elements associated with immune evasion. Significantly, chromatin accessibility correlates with DNA methylation and gene expression, providing an understanding of epigenetic regulation in cancer.

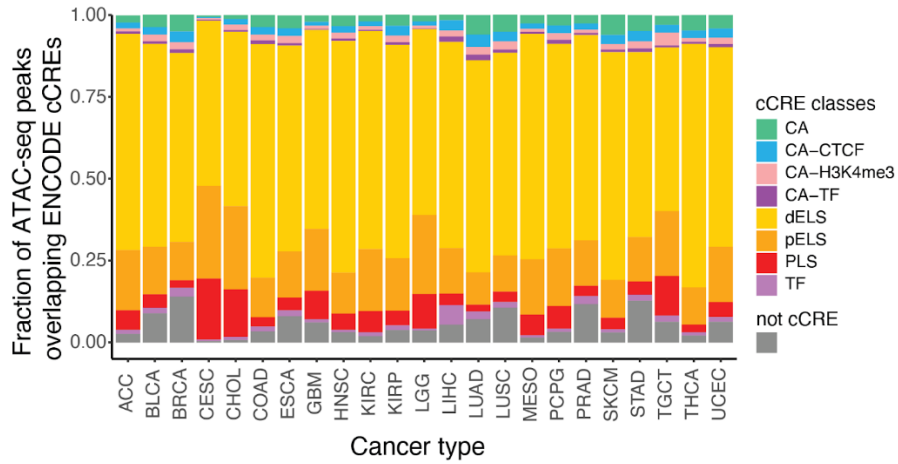


Figure 4: Fractions of ATAC-seq peaks overlapping cCREs in tumor samples in 23 cancer types

cCREs are from the ENCODE Registry. Each bar corresponds to a specific cancer type, with colors representing different cCRE classes, including chromatin accessible (CA), CA-CTCF, CA-H3K4me3, CA-transcription factor (TF), distal enhancers (dELS), proximal enhancers (pELS), promoters (PLS), and TF-only elements. Peaks that do not overlap with cCREs are indicated in gray. The high fraction of overlap across cancer types demonstrates the comprehensive nature of the ENCODE cCRE Registry in capturing regulatory elements relevant to cancer biology. Developed by the Moore Lab at UMass Chan Medical School.

Furthering the study by Corces et al., the Moore Lab at UMass Chan Medical School compared the ATAC-seq data from Corces et al. with the ENCODE cCREs. It determined that 93.7% of ATAC-seq peaks overlap with cCREs (Fig. 4). This high degree of overlap is significant as it suggests that the ENCODE cCRE Registry represents the active regulatory elements in cancer cells. Also, it is seen that the majority of the active cCREs are distal enhancers (dELS), proximal enhancers (pELS), and promoters (PLS). This indicates that these regulatory elements are active in these cancer types and could potentially drive cancer progression. This finding demonstrates the importance of cCREs in cancer gene regulation, as it indicates that many of the accessible chromatin regions in cancer cells are associated with known regulatory elements. (Zhang, Wang et al. 2024).

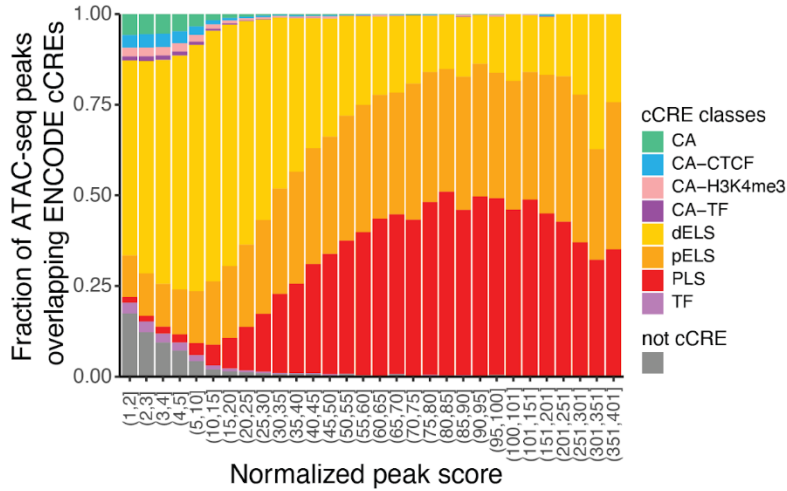


Figure 5: ATAC-seq peaks and cCRE overlap from tumor samples binned by ATAC-seq score

ATAC-seq peaks from tumor samples categorized into bins based on normalized peak scores, reflecting the intensity of ATAC-seq signals. It shows the fraction of peaks within each bin that overlap various cCRE classes. Each bar corresponds to a specific cancer type, with colors representing different cCRE classes, including chromatin accessible (CA), CA-CTCF, CA-H3K4me3, CA-transcription factor (TF), distal enhancers (dELS), proximal enhancers (pELS), promoters (PLS), and TF-only elements. Peaks that do not overlap with cCREs are indicated in gray. Lower-scoring peaks are non-overlapping, whereas higher-scoring peaks exhibit substantial overlap with specific cCRE classes. This analysis emphasizes the correlation between peak score and the likelihood of functional regulatory element overlap, highlighting the utility of peak scoring in identifying biologically relevant regions. Developed by the Moore Lab at UMass Chan Medical School.

Furthering the interpretation, the Moore Lab looked at cCRE distribution based on ATAC-seq score rather than cancer type. Figure 5 shows a correlation between normalized peak scores and the likelihood of overlap with regulatory features. Peaks with lower scores predominantly fall into the "not cCRE" category, indicating they are less likely to correspond to regulatory elements. In contrast, as peak scores increase, there is an increase in the proportion of peaks overlapping with cCREs, particularly those classified as promoter-like sequences (PLS) and enhancer elements (dELS and pELS). This suggests that high-confidence ATAC-seq peaks are more likely to represent biologically meaningful regulatory regions.

1.6 Goal of this Study

This study seeks to understand the role of cCREs in cancer development, including their function, distribution, overlap between 23 different types of cancer, and comparison to healthy tissues. The potential impact of this study is significant, as it will identify regulatory regions activated or altered in cancer, allowing for the understanding of the role of epigenetic changes

in gene regulation and expression. These findings could potentially lead to novel therapeutic strategies that target the dysregulated regulatory elements in cancer cells.

2. Methods and Results

This project aimed to determine the role of cCREs in cancer development. In a preliminary study, the Moore Lab at UMass Chan Medical School compared the ATAC-seq data from Corces et al. with the ENCODE cCREs (Abascal, Acosta et al. 2020) finding that 93.7% of ATAC-seq peaks overlap with cCREs. This finding hints at a potential relationship between cCREs and chromatin peaks, indicating the possible significance of cCREs in cancer development. The preliminary study highlighted the importance of studying cCREs to understand their function, distribution, cancer activity, and relatedness across 23 different types of cancer. To better understand the relationship hinted at during the Moore Lab preliminary study, a deeper look into the role of cCREs was performed. The computational pipeline in Figure 6 was utilized to understand the distribution of cCREs across patient and cancer type, how copy number variation and cCREs are correlated, and to enable an ontology analysis of cCREs' related genes and biological processes.

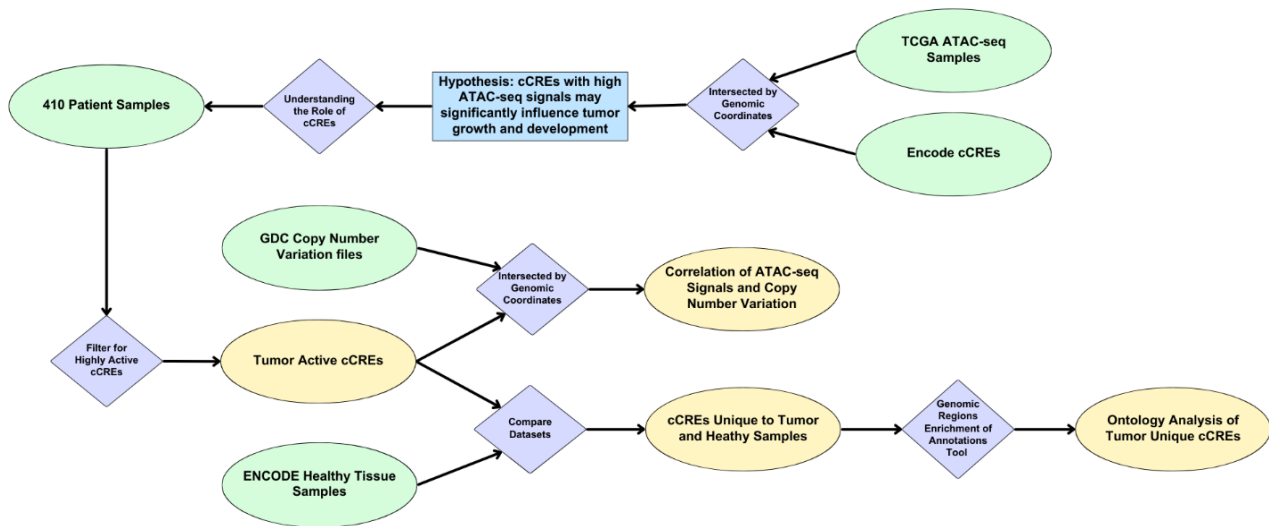


Figure 6: Computational pipeline built and executed during the study

Task methodology and results are detailed in the text. Task one resulted in the hypothesis in the blue rectangle. Task two resulted in the yellow oval titled “Tumor Active cCREs”. Task three resulted in the yellow circle titled “Correlation of ATAC-seq Signals and Copy Number Variation”. Task 4 resulted in the two yellow ovals titled “cCREs Unique to Tumor and Healthy Samples” and “Ontology Analysis of Tumor Unique cCREs”. Green ovals are inputs, purple diamonds are computational steps, and yellow circles are outputs.

2.1 Task 1: Reproducing ATAC-seq and ENCODE cCRE Plots From the Moore Lab

The two plots visualizing the relationship between ATAC-seq peaks and ENCODE cCREs created by the Moore Lab at UMass Chan Medical School (Figs. 3 and 4) had to be reproducible to determine if the plots were significant. The overlap of cancer-specific ATAC-seq peaks with ENCODE cCREs was analyzed to do this. The methodology combined data preprocessing, overlap calculation using Bedtools, classification of overlaps, and visualization of results. It was designed for reproducibility, modularity, and scalability, enabling a comprehensive understanding of the regulatory landscape across different cancer types.

2.1.1 Distribution of ATAC-seq Peaks Overlapping ENCODE cCREs by cCRE Class

To reproduce the first plot, the process began with filtering the peaks of the ATAC-seq data to select the peaks associated with the specific cancer type and keeping the first four columns of the file, which were the sequence names, start, end, and cancer type. This data was then written in a .bed file to execute the next step. During this step, using the tool Bedtools intersect, the filtered cancer peak file and a file of the ENCODE cCREs were compared, and a .bed file of the overlapped data was created (Figs. 7 and 8).

Peak Chromosome	Peak Start	cCRE End	Cancer Sample
chr2	112584205	112584706	ACC_10013
chr2	112596243	112596744	ACC_10016
chr1	21725692	21726193	ACC_1002
chr2	112713088	112713589	ACC_10025
chr2	112763935	112764436	ACC_10029

Figure 7: Filtered peaks of ATAC-seq data of Adenoid Cystic Carcinoma (ACC)

Peak Chromosome	Peak Start	Peak End	Cancer Sample	cCRE Chromosome	cCRE Start	cCRE End	rDHS ID	cCRE ID	cCRE Class
chr2	112541661	112542162	ACC_10008	chr2	112541897	112542087	EH38D5224555	EH38E3365367	PLS
chr2	112541661	112542162	ACC_10008	chr2	112542093	112542332	EH38D5224556	EH38E3365368	PLS
chr1	21673421	21673922	ACC_1001	chr1	21673379	21673727	EH38D2136805	EH38E1326356	dELS
chr2	112584205	112584706	ACC_10013	chr2	112584128	112584291	EH38D5224579	EH38E3365384	pELS
chr2	112584205	112584706	ACC_10013	chr2	112584317	112584667	EH38D5224580	EH38E3365385	PLS

Figure 8: Overlapped data using Bedtools intersect of Adenoid Cystic Carcinoma (ACC)

The intersected data were loaded, and specific columns were selected to focus on relevant information: the start and end coordinates of the peaks, the start and end coordinates of the cCRE regions, and the name of the cCRE region (cCRE ID) (Fig. 7).

Next, the function grouped the data by cCRE type to count overlaps for each class. It also identified peaks that overlap with multiple cCREs, counting these cases separately. Finally, the results are saved to two output files for each cancer type, one containing the counts of overlaps by cCRE type (Fig. 9) and another detailing chromatin peaks with multiple cCRE overlaps (Fig. 10).

```

cCRE_type    overlap_count
Chromatin Accessible    757
Chromatin Accessible + CTCF    625
Chromatin Accessible + H3K4me3    337
Chromatin Accessible + TF    227
Distal Enhancer    31312
Not cCRE    813
Promoter-like    2298
Proximal Enhancer    9278
TF only    431

```

Figure 9: Counts of overlaps by cCRE type of Adenoid Cystic Carcinoma (ACC)

```

multiple_overlap_count
13984

```

Figure 10: Multiple overlap count of Adenoid Cystic Carcinoma (ACC)

To better visualize the fraction of ATAC-seq peaks overlapping each cCRE type, a stacked bar chart was generated to show each of the 23 cancer types (Fig. 11). The chart highlighted differences in the regulatory landscape among cancer types, with the x-axis representing cancer types and the y-axis showing the fraction of peaks in each cCRE category. The bars were color-coded for clarity, with each cCRE class assigned a unique color (Fig. 11). This visualization step summarized the overlap analysis, making the results clear and easily understandable. The plot was identical to that produced by the Moore lab (Fig. 4).

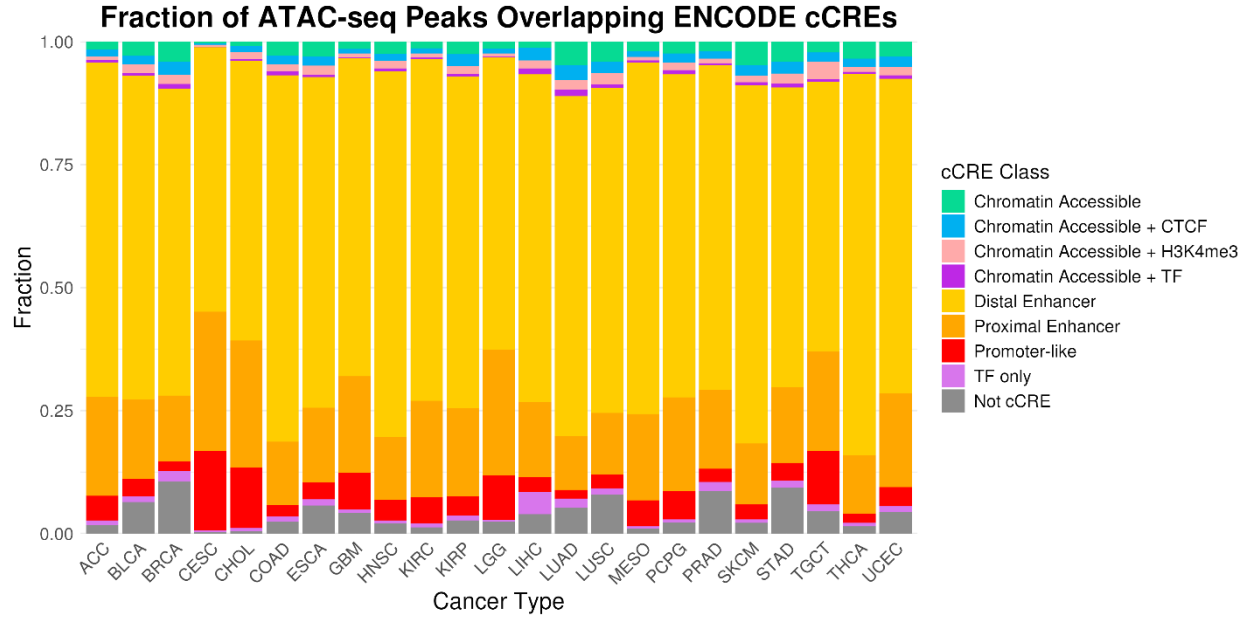


Figure 11: Reproduced plot showing fractions of ATAC-seq peaks in tumor samples in 23 cancer types.

2.1.2 Distribution of ATAC-seq Peaks Overlapping ENCODE cCREs by Peak Score

A similar pipeline was employed to reproduce the second plot. This pipeline analyzed overlaps between ATAC-seq peaks, grouped by normalized peak scores and ENCODE cCREs. By focusing on peak scores rather than cancer types, this approach offered insights into the regulatory landscape across a range of signal intensities. This is important because the higher the signal intensity, the more likely the region influences DNA transcription.

First, the ATAC-seq data are read into the data frame, and the peaks are filtered into groups within the specified score range. Next, the filtered peaks are stored in a .bed file that includes the chromosome, start, end, and cancer name (Fig. 12).

Peak Chromosome	Peak Start	cCRE End	Cancer Sample
chr8	86259159	86259660	BRCA_100005
chr8	86289547	86290048	BRCA_100011
chr8	86367044	86367545	BRCA_100038
chr8	86576967	86577468	BRCA_100078
chr8	86577468	86577969	BRCA_100079

Figure 12: Filtered peaks of ATAC-seq data of the 1 - 2 peak score range

With the newly created .bed file of the filtered peak score ranges, Bedtools intersect was employed to overlap the filtered peaks and the ENCODE cCREs, which output a .bed file of the combined data (Fig. 13).

Peak Chromosome	Peak Start	Peak End	Cancer Sample	cCRE Chromosome	cCRE Start	cCRE End	rDHS ID	cCRE ID	cCRE Class
chr8	86617975	86618476	BRCA_100085	chr8	86618018	86618289	EH38D6441446	EH38E4474301	dELS
chr8	86617975	86618476	BRCA_100085	chr8	86618441	86618672	EH38D5966028	EH38E3843529	dELS
chr8	87663868	87664369	BRCA_100108	chr8	87664014	87664166	EH38D5966319	EH38E4474482	CA-TF
chr8	87731961	87732462	BRCA_100109	chr8	87732032	87732375	EH38D5966340	EH38E3843659	dELS
chr1	120133709	120134210	BRCA_10011	chr1	120133652	120133828	EH38D2218411	EH38E1381789	dELS

Figure 13: Overlapped data using Bedtools intersect of the 1 - 2 peak score range

Using the intersect data, a function was created to count the overlaps for each cCRE type in each score range. This function outputs a file for each of the predefined peak score ranges, containing the number of identified cCREs for each of the cCRE types (Fig. 14)

```

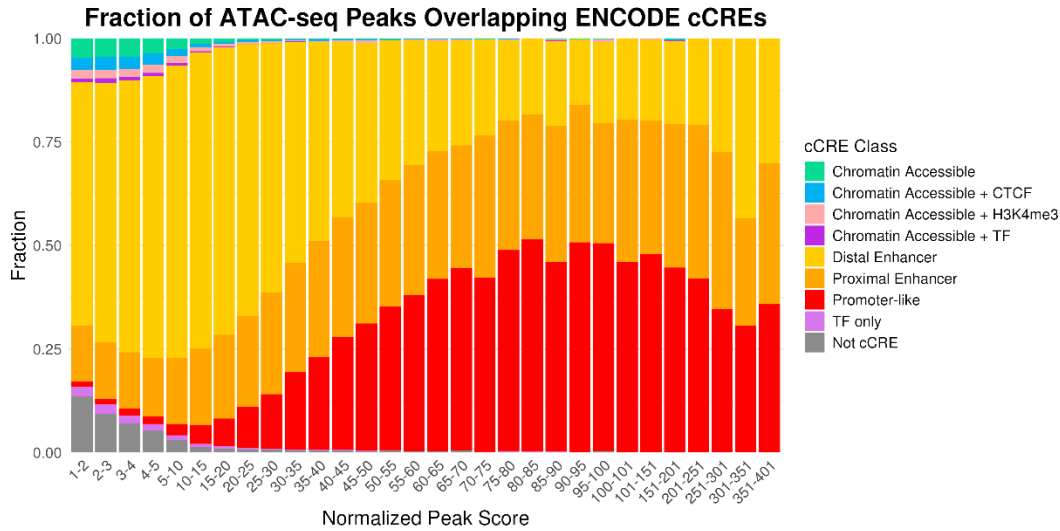
cCRE_type    overlap_count
Chromatin Accessible    1570
Chromatin Accessible + CTCF    946
Chromatin Accessible + H3K4me3    729
Chromatin Accessible + TF    281
Distal Enhancer    19487
Not cCRE    4471
Promoter-like    443
Proximal Enhancer    4460
TF only    792

```

Figure 14: Counts of overlaps by cCRE type of the 1 - 2 peak score range

These data allowed for the comparison of cCRE distributions across the score ranges. This was visualized using a stacked bar chart (Fig. 15). The stacked bar chart visualized the fraction of peaks overlapping each cCRE type across score ranges. The x-axis represented normalized peak score ranges, while the y-axis showed the fraction of peaks in each cCRE category. Bars were color-coded for clarity, with each cCRE class assigned a unique color. The plot

highlighted how the regulatory landscape varied across signal intensities (Fig. 15). It exactly reproduced the plot produced by the Moore lab (Fig. 5).



$$Z = (\log(\text{mean_signal}) - \mu) / \sigma$$

Figure 16: Z-score calculation formula

In the formula, μ and σ represent the mean and standard deviation of the log-transformed values, respectively. Regions with no signal were assigned a z-score of -10. The resulting z-scores were then filtered to retain only those regions overlapping with cCREs from the ENCODE Registry (GRCh38-cCREs.bed) using Bedtools intersect. This approach enabled a standardized comparison of accessibility across regulatory regions in diverse samples.

During clinical testing, patient samples are often duplicated to ensure that they were correctly taken and to remove bias. After the z-score algorithm, samples from the same patient were matched, and the z-score was averaged—this output 410 unique tumor samples, removing duplicate tumor samples from downstream analysis. Figure 17 depicts the structure of one of the bed files.

Chromosome	Start	End	cCRE ID	Z-score
chr1	10033	10250	EH38D4327497	1.597679
chr1	10385	10713	EH38D4327498	0.827120
chr1	16097	16381	EH38D6144701	-0.833801
chr1	17343	17642	EH38D6144702	0.025374
chr1	29320	29517	EH38D6144703	-0.666038
chr1	66350	66509	EH38D4327503	-10.000000
chr1	78990	79238	EH38D4327504	-5.199002
chr1	79304	79545	EH38D4327505	-5.583837
chr1	91087	91426	EH38D6144707	0.296110
chr1	98867	99117	EH38F0000001	-0.829871

Figure 17: Example of LUSC patient sample file after Z-score calculation.

2.2.2 Developing a cCRE Atlas of Highly Active cCREs

To determine which cCREs are significantly active, a pipeline was built to filter for Z-score. First, the directory with the 410 tumor samples and each cCRE with its Z-score was read into the script. Since the directory contained 410 individual files, each containing many cCREs (Fig. 17), this data had to be processed into one dataset for easier comparison and optimization. Each file was read into the data frame, and the column names were added. Since the column

names did not include the cancer type, a function was utilized to extract the cancer name from the file name and create a new column with the cancer name. To determine if a cCRE was highly active, a threshold of a z-score of 1.64 or greater was implemented. This means that at the 95% confidence interval, a cCRE is determined to be highly active (Fig. 18).

Chromosome	Start	End	cCRE ID	Z-score
chr1	181289	181639	EH38D4327525	2.669138
chr1	267925	268171	EH38D4327544	2.028651
chr1	778570	778919	EH38D4327580	3.834329
chr1	779023	779182	EH38D6144726	2.341875
chr1	779715	780047	EH38D6144727	2.306726
chr1	827417	827767	EH38D4327609	3.112002
chr1	869718	870065	EH38D4327653	2.025624
chr1	904578	904918	EH38D4327691	2.429117
chr1	921184	921360	EH38D4327731	1.769248
chr1	923719	924069	EH38D4327738	2.152781

Figure 18: Example of LUSC patient sample file only including Z-scores of 1.64 or greater

After this, the number of instances of each cCRE per cancer type determined to be highly active based on the thresholds was counted. This output is a .csv file containing the chromosome, start, end, cCRE ID, cancer type, number of entries, and the final count. (Fig. 19). The number of entries is the number of samples per cancer type in which the cCRE was present, and the final count is the number of cCREs that met the z-score threshold. The final count was used to create the histograms seen in Figure 20.

```
"chromosome","start","end","cCRE_id","cancer_type","n_entries","final_count"
"chr1",10033,10250,"EH38D4327497","ACCx",9,5
"chr1",10033,10250,"EH38D4327497","BLCA",10,4
"chr1",10033,10250,"EH38D4327497","BRCA",31,21
"chr1",10385,10713,"EH38D4327498","ACCx",9,1
"chr1",10385,10713,"EH38D4327498","BLCA",10,0
"chr1",10385,10713,"EH38D4327498","BRCA",31,0
"chr1",16097,16381,"EH38D6144701","ACCx",9,0
"chr1",16097,16381,"EH38D6144701","BLCA",10,0
"chr1",16097,16381,"EH38D6144701","BRCA",31,0
```

Figure 19: First nine rows of the cCRE Atlas

2.2.3 Organizing Data for All Cancers Overall

Histograms were created to visualize the count of active cCREs. To do this, an R script was used to read the cCRE Atlas and plot the corresponding data. The script combined all the data for each cancer type and determined the counts of each bar in the histogram by using the final count. For example, the first row of data in Figure 19 shows that cCRE EH38D4327497 for cancer type ACC was determined to be highly active in five samples. This means that five counts will be attributed to the fifth column on the Adenoid Cystic Carcinoma (ACC) histogram (Fig 20). Each count represents an ACC sample in which the cCRE was highly active. There were nine total ACC samples.

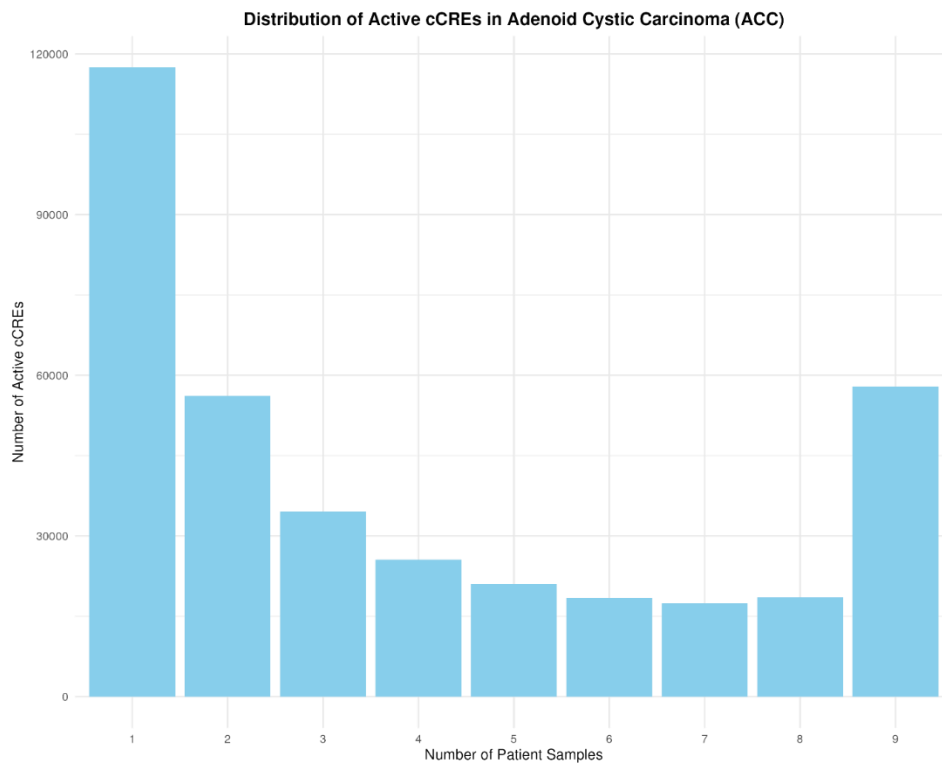


Figure 20: Histogram of Adenoid Cystic Carcinoma (ACC) active cCREs across patient samples

Figure 20 shows the distribution of highly active cCREs in ACC. The histogram shows that the most significant number of active cCREs is seen in lower numbers of samples. This could suggest that these cCREs play a substantial role in the sample they are present in, highlighting the differences between the same cancer in different patients. This suggests that although different patients can have the same cancer, there can be different cCREs that influence their cancer, highlighting the complexity of cancer and the need for individualized treatment.

The increased count of cCREs in the ninth bar, representing cCREs active in all nine ACC samples, likely reflects a substantial representation of elements crucial for maintaining essential cellular functions. In the study by Fan et al. (2021), the authors showed that a specific subset of ubiquitously accessible promoter-like cCREs (ubi-PLSs) is highly enriched in CpG dinucleotides and is predominantly associated with genes that are consistently expressed across a wide range of tissues. These ubi-PLSs recruit a network of ubiquitously expressed transcription factors and cofactors to ensure the robust, basal transcription required for cell viability (Fan, Moore et al. 2021).

2.2.4 Cross-Cancer Comparison of Highly Active cCREs

It is important to understand the relationship between each type of cancer and what active cCREs they share. This information can be used to understand if multiple cancer types are similar, which could result in similar treatments, or if a cancer type is very different, resulting in a targeted therapy. Using an R script, a binary matrix was created. The binary matrix determined the relatedness between cancers, based on the number of active cCREs they have in common. The binary matrix only focused on whether a specific cCRE was active in the cancer type, not the overall count of cCREs. This means that if the cCRE is scored as active in eight cancer samples, the value used in the matrix calculation is one, as it is a binary computation; zero means not active, and one means active. Using the pheatmap package in R, a heatmap was created to visualize the binary matrix.

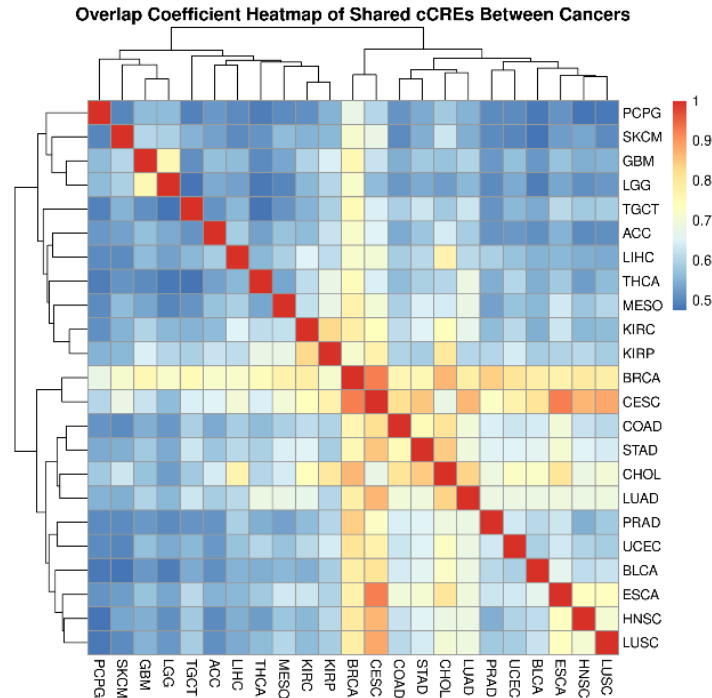


Figure 21: Overlap Coefficient Heatmap showing the relatedness of cancer types in terms of active cCREs

In the heatmap (Fig. 21), the relatedness of each cancer type in terms of active cCREs is seen. The diagonal red line of squares represents a comparison of each cancer type with itself, which naturally results in the maximum overlap coefficient of one. The red/orange cells indicate a high overlap coefficient, meaning the two cancer types share a significant proportion of their active cCREs. The yellow/blue cells indicate a lower overlap coefficient, meaning there is minimal sharing of active cCREs between the two cancer types. The rows and columns in the heatmap are clustered hierarchically to group cancer types with similar cCRE profiles. Close clusters suggest cancer types with similar regulatory landscapes. For example, cancers like BRCA (Breast Cancer) and CESC (Cervical Cancer) cluster together, suggesting they have similar active regulatory elements. BRCA and CESC also have higher overlap with many other cancers, suggesting they may have common regulatory landscapes, with fewer unique elements, perhaps representing a ‘baseline’ cancer phenotype. Also, some cancer types show minimal overlap, highlighting unique regulatory profiles that could point to specialized mechanisms or tumor-specific behaviors.

2.3 Task 3: Understanding Copy Number Variations and How cCREs are Related

The goal of task three was to understand how cCREs may play a role in copy number variation, a common theme of cancer. By comparing the ENCODE cCREs with CNV data

gathered from GDC, the role and degree that cCREs play in copy number variation were understood.

2.3.1 Preparing TCGA Tumor Files and GDC Copy Number Variations

To understand how cCREs can be related to copy number variations, the active cCREs from task two were compared to copy number variation (CNV) files. Derived from the Genomic Database Commons, CNV files are datasets that describe changes in the number of copies of genomic segments. These files are generated by analyzing genomic data and are used to identify regions of the genome that have been amplified or deleted. CNVs can vary widely in length, ranging from one kilobase to a few megabases. A copy number of two means that the sequence is normal, a copy number above two is an amplification, and a copy number below two is a deletion. Most copy numbers in cancer typically are zero to four, and in rare cases, ten or greater (Feuk, Carson et al. 2006). A JSON API query was built to download the CNVs efficiently. The query was built to access GDC and look for the 23 cancer types under the TCGA projects. Next, it retrieved all the related CNV files for those projects and downloaded them to the working directory. Once completed, all CNV files related to the 23 TCGA project cancers were ready to be used for analysis. Figure 22 shows a CNV file after downloading and sorting. The file contains many entries, each listing the chromosome, start, end, and copy number. In the end, one CNV file for each cancer type included all the known copy number changes related to that cancer.

Chromosome	Start	End	CNV
chrX	4682645	9102436	1
chrX	6675559	7341851	2
chrY	6762890	10051590	0
chrX	6903064	7064724	2
chrX	7066051	63552698	4
chrX	7336440	7711500	0

Figure 22: Sample data from a CNV file for Colon Adenocarcinoma (COAD)

2.3.2 Intersecting the CNVs and cCREs

Using a Bash script and Bedtools, the CNV and previously computed active cCRE files were compared. For each cancer sample, the start and end sequences of each entry in that file were compared with the CNV file of that type of cancer. If a cCRE sequence had start and end coordinates inside a CNV start and end, the copy number associated with the CNV file was assigned to that cCRE. This outputted a file for each cancer type that had cCRE entries linked to CNV entries, which shows if a cCRE is related to a region with a copy number change.

cCRE Chromosome	cCRE Start	cCRE End	cCRE ID	Z-score	CNV Chromosome	CNV Start	CNV End	CNV
chr1	778570	778919	EH38D4327580	3.397681	chr1	777382	783006	4
chr1	778570	778919	EH38D4327580	3.397681	chr1	690012	783006	0
chr1	778570	778919	EH38D4327580	3.397681	chr1	91515	973443	3
chr1	778570	778919	EH38D4327580	3.397681	chr1	95440	825042	3
chr1	778570	778919	EH38D4327580	3.397681	chr1	727034	970892	1
chr1	778570	778919	EH38D4327580	3.397681	chr1	758351	789489	4

Figure 23: Intersected CNV and cCRE file for Breast cancer (BRCA).

Figure 23 shows the initial intersect file, where it was observed that the same cCRE overlapped multiple CNV regions. Since the goal was to analyze each cCRE individually, it was necessary to consolidate these overlaps. To address this, a Bash script was used to scan through each file, and whenever the values in columns one through four matched, the copy number values were averaged. This ensured that each cCRE was represented by a single, average copy number value, as shown in Figure 24.

cCRE Chromosome	cCRE Start	cCRE End	cCRE ID	Z-score	CNV Chromosome	CNV Start	CNV End	CNV
chrX	152876921	152877220	EH38D6138903	2.860258	chrX	152871764	155699751	2
chrX	152878747	152879018	EH38D6138905	1.867047	chrX	152871764	155699751	2
chrX	152879022	152879357	EH38D4321291	2.827647	chrX	152871764	155699751	2
chrX	152879412	152879611	EH38D6138906	2.258287	chrX	152871764	155699751	2
chrX	152894336	152894655	EH38D6138919	3.083321	chrX	152871764	155699751	2
chrX	152895094	152895402	EH38D6472571	2.294389	chrX	152871764	155699751	2
chrX	152895493	152895694	EH38D6138922	2.107280	chrX	152871764	155699751	2
chrX	152895739	152896086	EH38D4321308	4.137757	chrX	152871764	155699751	2
chrX	152896190	152896344	EH38D6138923	3.229688	chrX	152871764	155699751	2
chrX	152896447	152896796	EH38D4321310	3.873270	chrX	152871764	155699751	2

Figure 24: Intersected CNV and cCRE file for Breast cancer (BRCA) with cCRE IDs concatenated

2.3.3 Computing the ATAC-seq Signal Value for Active cCREs

To understand the correlation of active cCREs and ATAC-seq signal intensity with association with CNVs, an algorithm was built to calculate the ATAC-seq value from the original 799 patient tumor samples. It was hypothesized that active cCREs with high ATAC-

seq scores might be associated with CNVs with greater change. To quantify chromatin accessibility across both regulatory and control regions, average signal intensities were computed from patient bigWig files using the tool bigWigAverageOverBed. For each file, the average ATAC-seq signal was calculated across two sets of genomic intervals: the ENCODE-defined candidate cis-regulatory elements (GRCh38-cCREs.bed) and a matched set of control regions (GRCh38-Anchors.bed). Like step 2.2.1, the duplicate samples were combined, and the ATAC-seq signal was averaged. After this, the value was transformed to a log2 ratio, which allowed for easier visualization due to high values. Figure 25 shows the resulting file.

cCRE ID	Number of Base Pairs	Number of Base Pairs Covered	ATAC-seq Signal
EH38D0001261	291	291	1.775487
EH38D0001472	314	314	1.353687
EH38D0002482	341	341	1.869331
EH38D0004420	223	223	0.497275
EH38D0004750	201	201	0.303829
EH38D0004910	233	233	-0.423832
EH38D0005391	334	334	1.300329
EH38D0006760	201	201	1.801256
EH38D0009140	201	201	0.668464
EH38D0011504	294	294	4.032233

Figure 25: ATAC-seq file for Bladder Urothelial Cancer (BCLA).

After preparing the ATAC-seq files, they were intersected with the previously intersected CNV and cCRE files, seen in Figure 24. To do this, a lookup table was created. If the value in column four of the CNV and cCRE merged files matches the value of column one in the ATAC-seq file, the ATAC-seq value was appended to the CNV and cCRE merged file. This output a file for each patient sample that had cCRE, CNV, and ATAC-seq signal, as seen in Figure 26.

cCRE Chromosome	cCRE Start	cCRE End	cCRE ID	Z-score	CNV Chromosome	CNV Start	CNV End	CNV CNV	ATAC-seq Signal
chrX	3262937	3263264	EH38D6088375	1.770141	chrX	3202098	5302236	4	3.738703
chrX	3271631	3271878	EH38D6088381	2.128499	chrX	3202098	5302236	4	4.297239
chrX	3271883	3272216	EH38D4256751	2.496404	chrX	3202098	5302236	4	4.865315
chrX	3286254	3286530	EH38D6088388	1.788673	chrX	3202098	5302236	4	3.772899
chrX	3345231	3345581	EH38D6088411	2.640663	chrX	3202098	5302236	4	5.079534
chrX	3345621	3345927	EH38D6088412	1.672939	chrX	3202098	5302236	4	3.587911
chrX	3345944	3346148	EH38D6088413	1.714344	chrX	3202098	5302236	4	3.711365
chrX	3346498	3346842	EH38D6088415	2.092188	chrX	3202098	5302236	4	4.232856
chrX	3346977	3347302	EH38D6462303	2.308392	chrX	3202098	5302236	4	4.567953
chrX	3363051	3363331	EH38D6088431	2.228080	chrX	3202098	5302236	4	4.453544

Figure 26: Intersected cCRE, CNV, and ATAC-seq file of Bladder Urothelial Cancer (BCLA)

2.3.4 Visualizing the Correlation of Highly Active cCREs and ATAC-seq Signal Intensity

To visualize the correlation between active cCREs and ATAC-seq signals, correlation plots were created. First, to prepare for plotting, the cCRE, CNV, and ATAC-seq files were cleaned to only include the necessary data. Also, the CNV values were transformed to a log2 ratio, to be consistent with the log2 ratio ATAC-seq signals and to better visualize larger values. Figure 27 shows the resulting file.

cCRE Chromosome	cCRE Start	cCRE End	Z-score	CNV	ATAC-seq Signal
chrX	3262937	3263264	1.770141	1	3.738703
chrX	3271631	3271878	2.128499	1	4.297239
chrX	3271883	3272216	2.496404	1	4.865315
chrX	3286254	3286530	1.788673	1	3.772899
chrX	3345231	3345581	2.640663	1	5.079534
chrX	3345621	3345927	1.672939	1	3.587911
chrX	3345944	3346148	1.714344	1	3.711365
chrX	3346498	3346842	2.092188	1	4.232856
chrX	3346977	3347302	2.308392	1	4.567953
chrX	3363051	3363331	2.228080	1	4.453544

Figure 27: Optimized file of intersected cCRE, CNV, and ATAC-seq data of Bladder Urothelial Cancer (BCLA)

To generate visualizations of copy number variation (CNV) and chromatin accessibility (ATAC-seq) across the genome, a Python script was executed in a SLURM-managed high-performance computing environment. The script performed a series of structured steps to process, visualize, and save genome-wide CNV and ATAC-seq signals. First, the script identified the input BED file corresponding to the assigned SLURM array task ID and derived

its associated cancer type from the directory structure. Chromosome sizes for the GRCh38 reference genome were defined, and chromosome offsets were computed to enable genome-wide coordinate plotting. Two vertically aligned matplotlib subplots were initialized: the top for CNV signal and the bottom for ATAC-seq signal.

Each BED file was read in chunks to efficiently handle large datasets. For each chunk, genomic midpoints were calculated and adjusted to genome-wide coordinates using chromosome offsets. CNV and ATAC-seq log2-transformed values were extracted from the input file and plotted against the genomic coordinates, with chromosome-specific color coding. CNV values were visualized as scattered points, while ATAC-seq signals were rasterized to optimize rendering performance for high-density data. Figure 28 shows the resulting plot.

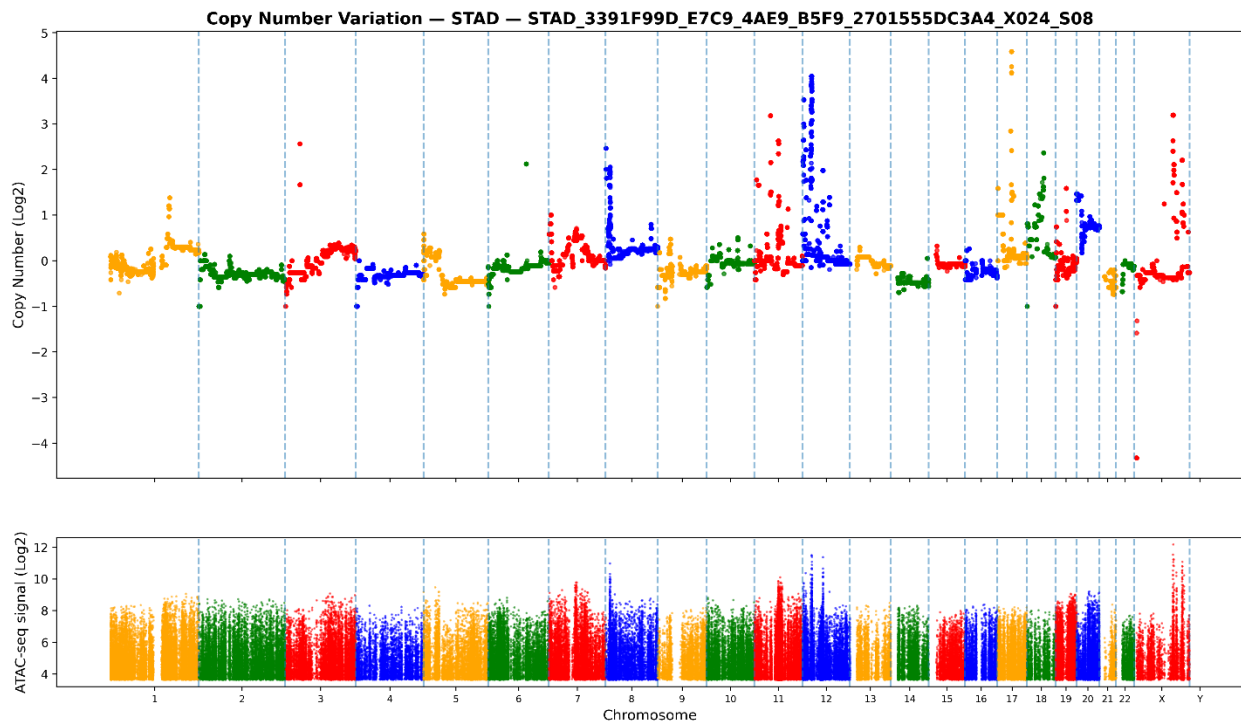


Figure 28: CNV and cCRE ATAC-Seq scores show correlation.

Derived from a patient with Stomach Adenocarcinoma (STAD). Top plot shows log2 ratio of CNV of active cCREs across all chromosomes. The bottom plot shows log2 ratio ATAC-seq signal intensity of active cCREs across all chromosomes.

Figure 28 shows the correlation between active cCREs and ATAC-seq signal in a single patient sample. The top plot is the CNV, with each point representing a cCRE and its value is the log2 change in copy number. A positive value is an amplification, and a negative value is a deletion, with zero being no CNV. The bottom plot is the log2 ATAC-seq signal intensity. Each

point represents a cCRE and higher values meaning a higher ATAC-seq signal, which corresponds to higher chromatin accessibility, a hallmark of DNA replication. In this sample, it is seen that areas of high CNV tend to have high ATAC-seq signals, which shows a correlation between these two factors. This is important because it suggests that cCREs may play a role in CNV, which shows a greater understanding of cancer dynamics and the significance of cCREs. Other patient samples followed this trend, but other samples did not. This highlights the complexity of the regulatory landscape and the need for personalized medicine.

2.4 Task 4: Comparison of Healthy and Tumor Active cCREs

The goal of task four was to compare the active cCREs seen in tumor and healthy datasets. Computational analyses using R, Python, and Bash showed the relationship between these datasets. Further analysis using Genomic Regions Enrichment of Annotations Tool (GREAT), allowed for an understanding of the genes and biological processes associated with the tumor active cCREs. These insights can lead to a stronger understanding of the role of cCREs in tumor function, which can help researchers develop targeted therapeutic approaches.

2.4.1 Preparing Unique cCRE Datasets for Healthy and Tumor Samples

The first step in comparing tumor and normal samples was preparing the datasets for computation. For the tumor dataset, using the previously generated cCRE Atlas, an R script was used to determine the presence of a cCRE ID. This was done by reading each line of the cCRE Atlas and determining presence based on column seven, which was the final count of active cCREs. If the value was greater than zero, the entry was copied to the new output file, seen in Figure 29. The script output a csv file containing all the active cCREs for each of the 23 cancer types.


```
"chromosome", "start", "end", "cCRE_id"
"chr1", 10033, 10250, "EH38D4327497"
"chr1", 10385, 10713, "EH38D4327498"
"chr1", 180743, 180904, "EH38D4327523"
"chr1", 181014, 181237, "EH38D4327524"
"chr1", 181289, 181639, "EH38D4327525"
"chr1", 184094, 184266, "EH38D6144712"
"chr1", 190809, 191058, "EH38D4327539"
"chr1", 191294, 191644, "EH38D4327540"
"chr1", 191759, 191935, "EH38D4327541"
```

Figure 29: Active tumor cCREs of Glioblastoma (GBM)

The normal dataset was downloaded from healthy tissue repositories on the ENCODE portal. The Weng Lab had previously downloaded these files, so the files were already ready to use. All the cCREs in the file were highly active, allowing for a direct comparison to the tumor dataset. Figure 30 shows the file structure. The cCRE ID in the fourth column was used throughout the task, as it was the matching ID format to the tumor dataset.

Chromosome	Start	End	cCRE ID	rDHS ID	cCRE Class
chr1	12601285	12601579	EH38D0011504	EH38E0070245	dELS
chr1	90721495	90721738	EH38D0076589	EH38E0102310	dELS
chr1	112466710	112466911	EH38D0092143	EH38E1067425	pELS
chr10	51700443	51700779	EH38D0217608	EH38E1082444	pELS
chr11	84800860	84801136	EH38D0344228	EH38E0009856	CA-H3K4me3

Figure 30: Healthy Brain cCREs derived from ENCODE

2.4.2 Comparison of Tumor, Healthy, and Shared cCREs

With both the input files ready for processing, the next step was to compare the cCREs in both datasets. Using a Bash script, the chromosome, start, end, and cCRE ID were read into the data frame. The script then looped through the input files and created three output files based on cCRE distribution. One file was cCREs unique to tumor, one file of cCREs unique to healthy, and one file of cCREs seen in both tumor and healthy datasets. Each of the files were structured in the same way, stating the chromosome, start, end, and cCRE ID (Figs. 31, 32, and 33).

Chromosome	Start	End	cCRE ID
chr1	90721495	90721738	EH38D0076589
chr1	112466710	112466911	EH38D0092143
chr10	51700443	51700779	EH38D0217608
chr13	44437980	44438325	EH38D0503113
chr14	32568059	32568304	EH38D0557832

Figure 31: Shared cCREs of Healthy Brain ENCODE data and Glioblastoma (GBM) Tumor Data.

Chromosome	Start	End	cCRE ID
chr11	84800860	84801136	EH38D0344228
chr11	114152896	114153099	EH38D0364947
chr13	110908022	110908284	EH38D0545618
chr14	51088843	51089044	EH38D0568948
chr14	58362826	58363167	EH38D0574558

Figure 32: Healthy cCREs of Healthy Brain ENCODE data and Glioblastoma (GBM) Tumor Data.

Chromosome	Start	End	cCRE ID
chr1	10033	10250	EH38D4327497
chr1	10385	10713	EH38D4327498
chr1	180743	180904	EH38D4327523
chr1	181014	181237	EH38D4327524
chr1	184094	184266	EH38D6144712

Figure 33: Tumor cCREs of Healthy Brain ENCODE data and Glioblastoma (GBM) Tumor Data.

To visualize the similarities and differences between the cCREs seen in tumor and healthy datasets, a Venn diagram was created. By inputting the three distinct files and using the R Eulerr package to create a proportional Venn diagram, the distribution of cCREs was visualized (Fig. 34).

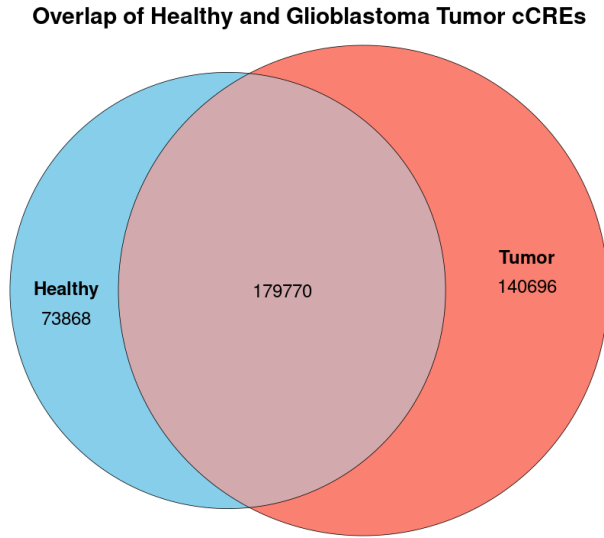


Figure 34: Proportional Venn diagram showing the overlap between cCREs of healthy Brain ENCODE data and Glioblastoma (GBM) tumor data

The Venn diagram depicted shows how many cCREs are seen in each dataset, tumor, healthy, and shared. It is seen that 179,770 active cCREs are shared between both datasets, meaning that there is lots of overlap between both datasets. Importantly, there are 140,696 cCREs that are highly active only in the tumor dataset. This group is important to look at, as they may have a large influence on tumor development. This was done in section 2.4.3.

2.4.3 Ontology Analysis of Tumor Active cCREs

As stated earlier, understanding the active cCREs distinct to the tumor set is important. Analysis of these cCREs can show the associated ontologies, revealing the associated genes and biological processes. To compute this, the Genomic Regions Enrichment of Annotations Tool (GREAT), developed by Stanford, was used (McLean, Bristor et al. 2010, Tanigawa, Dyer et al. 2022). GREAT operates by first defining extended regulatory domains around each gene, rather than simply linking a genomic region to the nearest gene. It assigns non-coding regions to these broader domains, capturing potential long-range regulatory interactions. Once the regions are associated with their potential target genes, GREAT performs statistical enrichment analyses to determine whether specific biological functions, pathways, or processes are overrepresented among the genes linked to the input regions compared to the genome at large.

GREAT needs two input files to perform the computation. The first is a background region, which is all the tumor and healthy data, and the second is a test region, which is only the tumor data. To prepare the background file, the tumor, healthy, and shared datasets were combined and converted to a bed file. GREAT allows for only the chromosome, start, and end, so the cCRE ID was removed. Figure 35 shows the resulting file that was used as the background region.

Chromosome	Start	End
chr1	12601285	12601579
chr11	84800860	84801136
chr11	114152896	114153099
chr13	110908022	110908284
chr14	51088843	51089044

Figure 35: Background file of cCREs of Glioblastoma (GBM)

To prepare the test file, the tumor dataset was copied, the cCRE ID column was removed, and turned into a bed file. Figure 36 shows the resulting test file.

Chromosome	Start	End
chr1	10033	10250
chr1	10385	10713
chr1	180743	180904
chr1	181014	181237
chr1	184094	184266

Figure 36: Test file of cCREs of Glioblastoma (GBM)

The next step was to run GREAT using these two files. After processing, the output files were copied to the working directory. The two output files that were of interest were the Ensembl Genes and the GO Biological Processes. To visualize the data, an R script was employed. The script read the two files and output a proportional world cloud of the Ensembl Genes (Fig. 37) and a bar plot of the top ten biological processes (Fig 38).

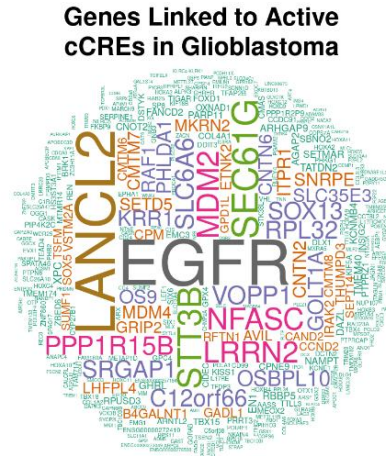


Figure 37: Proportional world cloud of genes associated with Glioblastoma (GBM) active cCREs

Figure 37 presents a proportional word cloud of genes linked to active cCREs in Glioblastoma (GBM). Each gene name corresponds to a gene significantly associated with at least one active cCRE, and the size of the label reflects the magnitude of association; larger labels represent higher statistical enrichment or frequency within the dataset.

The most prominent gene, EGFR (Epidermal Growth Factor Receptor), is a well-established oncogene in GBM. Amplification or mutation of EGFR, especially the EGFRvIII variant, drives tumor growth through enhanced cell proliferation, angiogenesis, and resistance to apoptosis (Libermann, Nusbaum et al. 1985). This finding aligns with prior studies indicating that EGFR is one of the most frequently altered genes in GBM, particularly in classical subtypes (Verhaak, Hoadley et al. 2010).

Another top-associated gene is LANCL2. Though less characterized in GBM, LANCL2 has been implicated in immune-metabolic regulation and may play a role in tumor microenvironment adaptation through modulation of reactive oxygen species and immune signaling pathways (Zhao, Zhou et al. 2021). Its prominence in this cCRE analysis suggests a potentially underexplored regulatory role in GBM progression or immune evasion.

Overall, this visualization highlights genes that are not only transcriptionally active but are also located near epigenetically active regulatory regions, reinforcing their functional importance in GBM biology. These gene-cCRE linkages provide a map of potentially

actionable targets and warrant further discussion in relation to tumorigenesis and therapy resistance mechanisms.

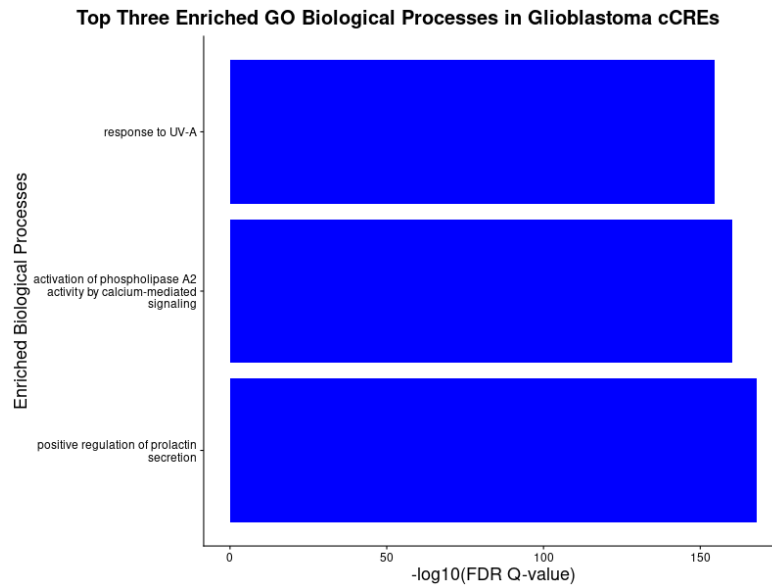


Figure 38: Bar plot of top three biological processes associated with Glioblastoma (GBM) active cCREs.

Figure 38 displays the top three significantly enriched Gene Ontology (GO) biological processes among cCREs active in Glioblastoma (GBM), as determined through gene set enrichment analysis. The processes shown, “response to UV-A, activation of phospholipase A2 activity by calcium-mediated signaling”, and “positive regulation of prolactin secretion”, highlight functional pathways that may be under regulatory control in GBM-specific chromatin landscapes.

The most enriched term, positive regulation of prolactin secretion, is particularly notable. Prolactin is traditionally recognized for its roles in lactation and reproductive biology. Recent evidence suggests that prolactin and its receptor (PRLR) may also influence tumorigenesis in several cancers, including Glioma. Prolactin signaling can activate downstream JAK2/STAT and MAPK pathways, which are known to promote cell proliferation, survival, and angiogenesis (Goffin, Bouchard et al. 1998). In GBM, overexpression of prolactin or PRLR has been associated with increased tumor cell proliferation and resistance to apoptosis. The presence of this GO term among enriched processes suggests that GBM-active cCREs may regulate genes involved in this pathway, offering a potential therapeutic target for disrupting hormone-responsive tumor signaling.

The enrichment of calcium-mediated activation of phospholipase A2 (PLA2) activity also provides mechanistic insight. PLA2 enzymes are involved in the release of arachidonic acid, a precursor for pro-inflammatory eicosanoids that modulate tumor microenvironments. Calcium signaling, a key regulator of PLA2 activity, has been implicated in GBM invasiveness and therapy resistance by promoting cytoskeletal remodeling and cell motility (Monteith, Davis et al. 2012).

Lastly, the response to UV-A pathway, though less direct in its relevance to brain tumors, may reflect shared DNA damage response mechanisms or oxidative stress pathways activated in GBM. UV-A radiation induces reactive oxygen species (ROS), and GBM cells are known to co-opt ROS signaling to support growth under hypoxic and inflammatory conditions (Diebold and Chandel 2016).

Together, these enriched biological processes highlight regulatory themes in GBM such as hormonal signaling, lipid-mediated inflammation, and oxidative stress response, all critical to tumor maintenance and progression.

3. Discussion

This study builds directly on the foundational work of the Moore Lab at UMass Chan Medical School and the chromatin accessibility mapping of Corces et al., further advancing the understanding of cis-regulatory element dynamics in cancer. By integrating high-resolution ATAC-seq datasets, comprehensive ENCODE cCRE annotations, and TCGA-derived copy number variation data, this research not only reproduces key findings from these earlier studies but also extends them with new insights into the landscape and functional impact of regulatory elements in tumor biology.

The Moore Lab and Corces et al. first demonstrated that a vast majority of ATAC-seq peaks in human cancers overlap known ENCODE cCREs, confirming that cancer cells extensively exploit the regulatory genome. This finding was reproduced and confirmed, revealing that enhancer-like and promoter-like elements are particularly prominent among accessible regions in diverse cancer types. Importantly, analysis of normalized peak scores indicated that higher ATAC-seq signal intensities are closely associated with a greater likelihood of functional

regulatory element overlap, highlighting the practical utility of peak scoring in prioritizing regions of biological interest.

By assembling a comprehensive cCRE atlas for 23 cancer types and rigorously filtering for high-confidence, highly active elements (using standardized z-score thresholds), this study systematically mapped the distribution of active cCREs both within and across cancer types. A notable finding was the substantial heterogeneity observed even within the same cancer type: most active cCREs are unique to individual tumor samples, underscoring the high degree of epigenetic individuality present in cancer and the likely need for personalized therapeutic approaches. However, the existence of a set of cCREs that are consistently active across samples and even across cancer types suggests that core regulatory circuits, such as those related to ubiquitously accessible promoters, are fundamental to tumor cell survival, supporting findings from Fan et al. (2021). The cross-cancer overlap heatmap further revealed clusters of cancers, such as BRCA and CESC, with similar cCRE activity profiles, which may open doors to shared therapeutic strategies for tumors with convergent regulatory features.

Integrating CNV data with cCRE and ATAC-seq information provided another critical layer of insight. The observed correlation between regions of high chromatin accessibility and CNV amplifications suggests that structural genomic alterations may preferentially target or co-opt regulatory regions, further driving oncogene expression and tumorigenesis. This underscores the evolving view that CNVs, beyond being structural markers, may play active functional roles in the reconfiguration of cancer regulatory networks.

When comparing tumor and healthy tissue cCRE profiles, the analysis revealed a large cohort of regulatory elements uniquely active in tumors, many of which are associated with genes and pathways previously implicated in cancer progression, including hormonal signaling, inflammation, and oxidative stress response. Ontology enrichment analyses, such as the association with prolactin signaling and calcium-mediated phospholipase activity, suggest that tumor-specific cCREs may orchestrate key processes contributing to malignancy and therapy resistance. The analysis also revealed many unknown genes and pathways, giving new insights into cancer development. These results collectively reinforce the concept that cCREs

are central players in the epigenetic regulation of cancer and provide a catalog of potential therapeutic targets.

However, several limitations remain. Biological variability within both tumor and healthy cohorts may confound interpretation, and the assignment of distal regulatory elements to their true gene targets is still imperfect without experimental chromatin conformation data. Additionally, while statistical enrichment is compelling, functional validation remains necessary to establish causality for individual regulatory elements.

Future research should focus on investigating specific cCREs at the molecular level to elucidate how their unique structures and regulatory functions contribute to tumor growth and progression. This could involve targeted experimental studies, such as CRISPR-mediated perturbations or reporter assays, to directly assess the functional consequences of altering particular cCREs in cancer cell lines and model systems. A deeper comparative analysis of cCRE activity and sequence between individual patients and across diverse cancer types will also be crucial for uncovering both shared regulatory mechanisms and patient-specific variations that may underpin therapeutic response or disease outcome. Integrating these findings with an expanded literature review will further clarify the roles of cCREs in modulating biological processes and gene networks relevant to oncogenesis, ultimately providing a foundation for developing precision medicine strategies that target the regulatory genome in cancer.

In summary, this work not only substantiates the earlier findings from the Moore Lab and Corces et al. but also provides new frameworks and resources that will support ongoing efforts to decode the regulatory genome in cancer. These advances move the field closer to a future where epigenetic profiling can directly inform precision oncology and the design of targeted epigenetic therapies.

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